CST1510 Programming for data communication and networks

Week 3: List, Multidimensional Lists, Functions and Lambda Expressions,

LAB

Welcome to Week 3 Python Lab Session!

This week, we will focus on the following key topics:

- **Slicing in lists**
- **Defining and using functions**
- **Lambda expressions for concise coding**
- **Working with multidimensional lists (nested lists)**

Once you've completed the tasks, choose one or two activities that demonstrate what you've learned this week to include in your portfolio.

Your portfolio should highlight your understanding of:

- Lists
- Functions
- Lambda expressions
- Multidimensional lists

Choose wisely!

ACTIVITY:

Example Activity 1

Write a Converter which would allow conversion from Fahrenheit to Celsius or from Celsius to Fahrenheit.

Conversion Formulas:
Fahrenheit = (Celsius * 1.8) + 32
Celsius = (Fahrenheit - 32)/1.8

In [ ]:
```python
# solution to activity 1

def cel_to_fah(celsius):
    """Calculate the value of fahrenheit from a giving number of celsius"
    fahrenheit = (celsius * 1.8) + 32.0
    return fahrenheit

def fah_to_cel(fahrenheit):
    """Calculate the value of celsius from a giving number of fahrenheit"
    celsius = (fahrenheit - 32.0) / 1.8
    return celsius


def main():
    """Main menu allowing choice between two functions """

    text = 'Enter;\n1 - to convert from celsius to fahrenheit\n' \
           '2 - to convert from fahrenheit to celsius\n'
    user_choice = input(text)

    if user_choice == '1':
        number = float(input('Enter the number of celsius > '))
        result = cel_to_fah(number)
        print(f'{number} celsius is {result} fahrenheit')

    elif user_choice == '2':
        number = float(input('Enter the number of fahrenheit > '))
        result = fah_to_cel(number)
        print(f'{number} fahrenheit is {result} celsius')

# Calling the main function

main()
```

In [ ]:
```python
#  solution to activity 1  - shorter version
def cel_to_fah(celsius):
    """Calculate the value of fahrenheit from a giving number of celsius"
    return (celsius * 1.8) + 32.0

def fah_to_cel(fahrenheit):
    """Calculate the value of celsius from a giving number of fahrenheit"
    return (fahrenheit - 32.0) / 1.8

def text_change(user):
    """ Helper function returning celsius, fahrenheit or other way around
    if user == '1':
        return 'celsius', 'fahrenheit'
    elif user == '2':
        return 'fahrenheit','celsius'


def main():
    """Main menu allowing choice between two functions """

    text = 'Enter;\n1 - to convert from celsius to fahrenheit\n' \
           '2 - to convert from fahrenheit to celsius\n'
    user_choice = input(text)
    v1, v2 = text_change(user_choice)
```

```
    number = float(input('Enter the number of {v1} > '))

    if user_choice == '1':
        result = cel_to_fah(number)

    elif user_choice == '2':
        result = fah_to_cel(number)

    print(f'{number} {v1} is {result:.2f} {v2}')

# Calling the main function
main()
```

Activity 2

Suppose we want to write a function to find the largest number, but we don't know the number of values.

We can create a function with variable-arguments length.

In [ ]:
```
# fsolution to activity 2
def find_max_number(*args):
    max_ = 0
    for i in args:
        if(i>max_):
            max_=i
    # return max number instead of printing it.
    return max_
```

In [ ]:
```
print("Max number is : ",find_max_number(10,5,15))
```

In [ ]:
```
print("Max number is : ",find_max_number(50,100,500,222,999,12,5))
```

In [ ]:
```
## Calling find_max_number() without any argument will return '0' because
## Otherwise, it would have generated an error since we're not handling w

print("Max number is : ",find_max_number())
```

Activity 3

Suppose we want to create a dynamic, user-input based, simple addition calculator.

The program will ask the user to keep entering the values to sum, and once done, just press ENTER key without entering a value to see the total.

In [ ]:
```
# Solution to Activity 3 - Simple recursive activity
def user_input_sum(n=0):

    user_input = input("Enter a number or hit ENTER key without a \
                        value to calculate total: ")

    if not user_input:
        return n
    else:
        n += int(user_input);
```

```
    return user_input_sum(n) # and called itself

print(f"\nThe total is: {user_input_sum()}")
```

Threshold:

1. Quiz

Suppose you want to develop a program to help a first grader practice addition, subtraction, multiplication, division, exponent or reminder. The user should have option of choosing which mathematical operation they wish to practice. The program randomly generates two single-digit integers, number1 and number2 and of course the user will provide answer to the question.

The program should include

- The applications should include a menu allowing the student to choose the
- operation they want to practice (ideally included it in function names of main).
- Use functions for each of the mathematical operations
- There should be a score calculating correct answers
- Consider printing some errors in situation when 'division by zero'
- Add functionality of keep practicing until the user decide otherwise.

In [ ]:

```

```

2.Write a program that reads integers from the user and stores them in a list.

Use 0 as a sentinel value to mark the end of the input. Once all of the values have been read your program should display them (except for the 0) in reverse order, with one value appearing on each line.

In [ ]:

```

```

3.Create a list of seven dice throws

The throws should be randomly generated and add to the list The list should be shuffle and print it out Sort the list descending order and print it out Count the occurrences and print them out

In [ ]:

```

```

4.Write a function that returns the sum of all the elements in a specified row in a matrix using the following header: def sum_row(matrix, rowIndex): Write a test program that reads a 4-by-4 matrix and displays the sum of each row.

In [ ]:

```

```

Typical:

5. Create a Scientific Calculator imitating the Fig1.Unit Conversion Chart

Allow the user to choose between length, temper, mass, volume, pressure, and energy. Create separate functions for each calculation within the the types of unit. Allow the user to choose between units. For example, if the user chooses length, then the programme should allow the user to choose between conversion: m/ft, in/cm, or mi/km. The user should then enter the value to calculate. Finely, the system should do the conversion and print out the result.

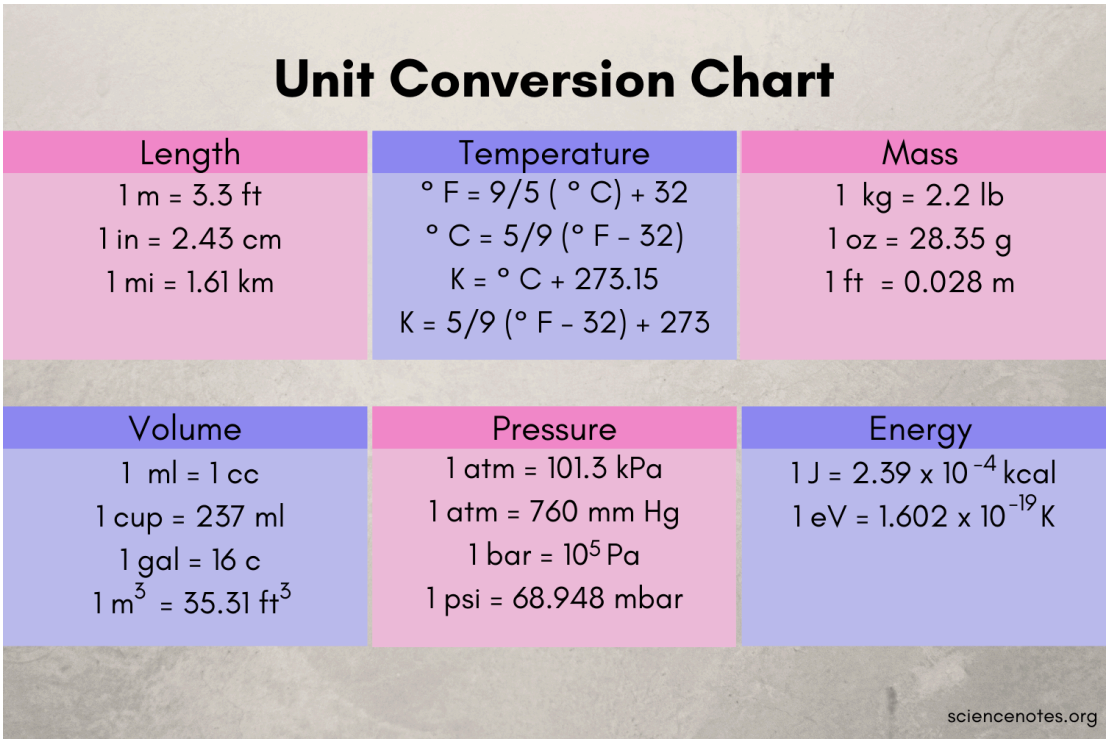

Fig1. Unit Conversion Chart

In [ ]:

6. Palindrome Checker

Write a function to check if a given string is a palindrome (reads the same backward as forward).

For example:

Some of the palindrome words are civic, radar, level, rotor, kayak, madam, refer, rotator, deified, racecar and reviver.

The function should one string, and return True it is palindrome, otherwise False.

In [ ]:

7. Text compression

Write a Python function called compress_string that takes a string as input and returns a compressed version of the string. The compression should replace repeated characters with the character followed by the count of repetitions. If the compressed string is longer than the original string, return the original string.

```
compressed = compress_string("aaabbbbcccddeee")
# The expected output is "a3b4c3d2e3" because 'a' is
repeated 3 times, 'b' 4 times, 'c' 3 times, etc.

compressed = compress_string("Hello")
# The expected output is "Hello", because the compressed
string in this case H1e1l2o1 is longer than the original
string "Hello"
```

In [ ]:

8.In order to win the top prize in a particular lottery, one must match all 6 numbers on his or her ticket to the 6 numbers between 1 and 49 that are drawn by the lottery organizer. Write a program that generates a random selection of 6 numbers for a lottery ticket. Display the numbers in ascending order.

In [ ]:

9.Python's standard library includes a method named count that determines how many times a specific value occurs in a list. In this exercise, you will create a new function named countRange which determines and returns the number of elements within a list that are greater than or equal to some minimum value and less than some maximum value.

Your function will take three parameters: the list, the minimum value and the maximum value. It will return an integer result.

Include a main program that demonstrates your function for several different lists, minimum values and maximum values. Ensure that your program works correctly for both lists of integers and lists of floating-point numbers. (For clarification.

Count the number of elements in a list that are greater than or equal to some minimum value and less than some maximum value.)

In [ ]:

Excellent:

10. Rewrite the quiz or the scientific converter including meanly lambda functions.

In [ ]:

11.A sublist is a list that makes up part of a larger list. A sublist may be a list containing a single element, multiple elements, or even no elements at all. For example, `[1], [2], [3]` and `[4]` are all sublists of `[1, 2, 3, 4]`. The list `[2, 3]` is also a sublist of `[1, 2, 3, 4]`, but `[2, 4]` is not a sublist `[1, 2, 3, 4]` because the elements `2` and `4` are not adjacent in the longer list. The empty list is a sublist of any list. As a result, `[]` is a sublist of `[1, 2, 3, 4]`. A list is a sublist of itself, meaning that `[1, 2, 3, 4]` is also a sublist of `[1, 2, 3, 4]`.

In this exercise you will create a function, isSublist, that determines whether or not one list is a sublist of another. Your function should take two lists, larger and smaller, as its only parameters. It should return True if and only if smaller is a sublist of larger. Write a main program that demonstrates your function.

In [ ]:
```
```